

数值分析第十七次作业

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1. 定义离散正弦变换 (DST) 为

$$F_k = \sum_{j=1}^{N-1} f_j \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leq k \leq N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \sum_{k=1}^{N-1} F_k \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leq j \leq N-1.$$

解答.

将 DST 代入可得

$$\begin{aligned} & \frac{2}{N} \sum_{k=1}^{N-1} F_k \sin\left(\frac{jk\pi}{N}\right) \\ &= \frac{2}{N} \sum_{k=1}^{N-1} \sum_{\ell=1}^{N-1} f_\ell \sin\left(\frac{\ell k\pi}{N}\right) \sin\left(\frac{jk\pi}{N}\right) \\ &= \frac{2}{N} \sum_{\ell=1}^{N-1} f_\ell \sum_{k=1}^{N-1} \sin\left(\frac{\ell k\pi}{N}\right) \sin\left(\frac{jk\pi}{N}\right) \\ &= \frac{1}{N} \sum_{\ell=1}^{N-1} f_\ell \sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) - \cos\left(\frac{(\ell+j)k\pi}{N}\right) \right). \end{aligned}$$

再注意到 $\forall t \in \mathbb{Z}$ 满足 $t/2N \notin \mathbb{Z}$, 都有

$$\sum_{k=1}^{N-1} \cos\left(\frac{tk\pi}{N}\right) = \frac{1}{2 \sin\left(\frac{t\pi}{2N}\right)} \left(\sin\left(\frac{(N-\frac{1}{2})t\pi}{N}\right) - \sin\left(\frac{\frac{1}{2}t\pi}{N}\right) \right) = \frac{1}{2}((-1)^{t-1} - 1),$$

故

(1) 当 $\ell = j$ 时,

$$\sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) - \cos\left(\frac{(\ell+j)k\pi}{N}\right) \right) = N - 1 - \frac{1}{2}((-1)^{2j-1} - 1) = N;$$

(2) 当 $\ell \neq j$ 时,

$$\sum_{k=1}^{N-1} \left(\cos \left(\frac{(\ell-j)k\pi}{N} \right) - \cos \left(\frac{(\ell+j)k\pi}{N} \right) \right) = \frac{1}{2}((-1)^{\ell-j-1} - 1) - \frac{1}{2}((-1)^{\ell+j-1} - 1) = 0;$$

从而 $\forall 1 \leq j \leq N-1$, 都有

$$\frac{2}{N} \sum_{k=1}^{N-1} F_k \sin \left(\frac{jk\pi}{N} \right) = \frac{1}{N} \sum_{\ell=1}^{N-1} f_{\ell} \sum_{k=1}^{N-1} \left(\cos \left(\frac{(\ell-j)k\pi}{N} \right) - \cos \left(\frac{(\ell+j)k\pi}{N} \right) \right) = \frac{1}{N} f_j \cdot N = f_j.$$

□

2. 定义离散余弦变换 (DCT) 为

$$F_k = \frac{1}{2}(f_0 + (-1)^k f_N) + \sum_{j=1}^{N-1} f_j \cos \left(\frac{jk\pi}{N} \right), \quad \forall 0 \leq k \leq N,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \left(\frac{1}{2}(F_0 + (-1)^j F_N) + \sum_{k=1}^{N-1} F_k \cos \left(\frac{jk\pi}{N} \right) \right), \quad \forall 0 \leq j \leq N.$$

解答.

将 DCT 代入可得

$$\begin{aligned} & \frac{2}{N} \left(\frac{1}{2}(F_0 + (-1)^j F_N) + \sum_{k=1}^{N-1} F_k \cos \left(\frac{jk\pi}{N} \right) \right) \\ &= \frac{2}{N} \left\{ \frac{1}{2} \left[\frac{1}{2}(f_0 + f_N) + \sum_{\ell=1}^{N-1} f_{\ell} + (-1)^j \left(\frac{1}{2}(f_0 + (-1)^N f_N) + \sum_{\ell=1}^{N-1} f_{\ell}(-1)^{\ell} \right) \right] \right. \\ & \quad \left. + \sum_{k=1}^{N-1} \left[\frac{1}{2}(f_0 + (-1)^k f_N) + \sum_{\ell=1}^{N-1} f_{\ell} \cos \left(\frac{\ell k\pi}{N} \right) \right] \cos \left(\frac{jk\pi}{N} \right) \right\} \\ &= \frac{2}{N} \left\{ \sum_{\ell=1}^{N-1} f_{\ell} \left[\sum_{k=1}^{N-1} \cos \left(\frac{\ell k\pi}{N} \right) \cos \left(\frac{jk\pi}{N} \right) + \frac{1}{2} + \frac{1}{2}(-1)^{j+\ell} \right] \right. \\ & \quad + f_0 \left[\frac{1}{2} \sum_{k=1}^{N-1} \cos \left(\frac{jk\pi}{N} \right) + \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2}(-1)^j \right) \right] \\ & \quad \left. + f_N \left[\frac{1}{2} \sum_{k=1}^{N-1} (-1)^k \cos \left(\frac{jk\pi}{N} \right) + \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2}(-1)^{N+j} \right) \right] \right\} \\ &= \frac{2}{N} \left(\sum_{\ell=1}^{N-1} f_{\ell} S_{\ell j} + \frac{1}{2} f_0 S_{0j} + \frac{1}{2} f_N S_{Nj} \right), \end{aligned}$$

其中 $\forall 0 \leq \ell, j \leq N$,

$$\begin{aligned} S_{\ell j} &= \sum_{k=1}^{N-1} \cos\left(\frac{\ell k \pi}{N}\right) \cos\left(\frac{j k \pi}{N}\right) + \frac{1}{2} + \frac{1}{2}(-1)^{j+\ell} \\ &= \frac{1}{2} \left[\sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell+j)k\pi}{N}\right) + \cos\left(\frac{(\ell-j)k\pi}{N}\right) \right) + 1 + (-1)^{j+\ell} \right]. \end{aligned}$$

类似第 1 题分类讨论可得

$$S_{\ell j} = \begin{cases} 0, & \ell \neq j; \\ \frac{N}{2}, & \ell = j \neq 0, N; \\ N, & \ell = j = 0, N; \end{cases}$$

故 $\forall 0 \leq j \leq N$, 都有

$$\frac{2}{N} \left(\frac{1}{2}(F_0 + (-1)^j F_N) + \sum_{k=1}^{N-1} F_k \cos\left(\frac{j k \pi}{N}\right) \right) = \frac{2}{N} \left(\sum_{\ell=1}^{N-1} f_{\ell} S_{\ell j} + \frac{1}{2} f_0 S_{0j} + \frac{1}{2} f_N S_{Nj} \right) = f_j.$$

□

3. 定义离散半波余弦变换 (Discrete Quarter-Wave Cosine Transform) 为

$$F_k = \sum_{j=0}^{N-1} f_j \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right), \quad \forall 0 \leq k \leq N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \left(\frac{1}{2} F_0 + \sum_{k=1}^{N-1} F_k \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right) \right), \quad \forall 0 \leq j \leq N-1.$$

解答.

将离散半波余弦变换代入可得

$$\begin{aligned} & \frac{2}{N} \left(\frac{1}{2} F_0 + \sum_{k=1}^{N-1} F_k \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right) \right) \\ &= \frac{2}{N} \left(\frac{1}{2} \sum_{\ell=0}^{N-1} f_{\ell} + \sum_{k=1}^{N-1} \sum_{\ell=0}^{N-1} f_{\ell} \cos\left(\frac{(\ell + \frac{1}{2})k\pi}{N}\right) \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right) \right) \\ &= \frac{2}{N} \sum_{\ell=0}^{N-1} f_{\ell} \left(\frac{1}{2} + \sum_{k=1}^{N-1} \cos\left(\frac{(\ell + \frac{1}{2})k\pi}{N}\right) \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right) \right) \\ &= \frac{1}{N} \sum_{\ell=0}^{N-1} f_{\ell} \left[1 + \sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) + \cos\left(\frac{(\ell+j+1)k\pi}{N}\right) \right) \right]. \end{aligned}$$

类似第 1 题分类讨论可得

$$\sum_{k=1}^{N-1} \left(\cos \left(\frac{(\ell-j)k\pi}{N} \right) + \cos \left(\frac{(\ell+j+1)k\pi}{N} \right) \right) = \begin{cases} -1, & \ell \neq j; \\ N-1, & \ell = j; \end{cases}$$

故 $\forall 0 \leq j \leq N-1$, 都有

$$\begin{aligned} & \frac{2}{N} \left(\frac{1}{2} F_0 + \sum_{k=1}^{N-1} F_k \cos \left(\frac{(j+\frac{1}{2})k\pi}{N} \right) \right) \\ &= \frac{1}{N} \sum_{\ell=0}^{N-1} f_{\ell} \left[1 + \sum_{k=1}^{N-1} \left(\cos \left(\frac{(\ell-j)k\pi}{N} \right) + \cos \left(\frac{(\ell+j+1)k\pi}{N} \right) \right) \right] \\ &= \frac{1}{N} f_j (1 + (N-1)) = f_j. \end{aligned}$$

□

4. 通过离散半波余弦变换推导矩阵

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -1 \end{pmatrix}_{N \times N}$$

的特征分解形式.

解答.

设 $\mathbf{x} = (x_0, x_1, \dots, x_{N-1})^T \in \mathbb{R}^N \setminus \{\mathbf{0}\}$ 满足 $\mathbf{Ax} = \lambda \mathbf{x}$, 则

$$\begin{cases} x_{j-1} - 2x_j + x_{j+1} = \lambda x_j, & \forall 1 \leq j \leq N-2; \\ -x_0 + x_1 = \lambda x_0; \\ x_{N-2} - x_{N-1} = \lambda x_{N-1}. \end{cases}$$

再引入虚节点 $x_{-1} = x_0, x_N = x_{N-1}$, 则

$$\begin{cases} x_{j-1} - 2x_j + x_{j+1} = \lambda x_j, & \forall 0 \leq j \leq N-1; \\ x_{-1} = x_0; \\ x_N = x_{N-1}. \end{cases}$$

$\forall 0 \leq k \leq N-1$, 取离散半波余弦变换的基向量 $\mathbf{x}^{(k)} \in \mathbb{R}^N$, 同时也引入虚节点, 即

$$x_j^{(k)} = \cos \left(\frac{(j+\frac{1}{2})k\pi}{N} \right), \quad \forall -1 \leq j \leq N,$$

直接代入上述方程组计算可得

$$\lambda_k = 2 \cos \left(\frac{k\pi}{N} \right) - 2 = -4 \sin^2 \left(\frac{k\pi}{2N} \right),$$

由此可得矩阵 $\mathbf{A}$ 的所有特征值和对应的特征向量.

□